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Cloud-Based ERP vs On-Premise ERP: Future Directions for Enterprises

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Executive Summary

This report compares cloud-based and on-premise Enterprise Resource Planning (ERP) systems in order to establish their long-term viability in various business situations. On-premise ERP is deployed locally, enabling maximum control, customization and data security but demands significant investment in infrastructure, qualified IT personnel and maintenance time. Meanwhile, cloud-based ERP uses vendor-managed cloud infrastructure to provide real-time access, scalability, accelerated implementation and lower upfront expenses, making it appealing to enterprises with limited resources. The key findings from case studies emphasize these differences: Engineering Company (EngCo) effectively embraced cloud-based ERP to eliminate data delays and improve cross-department efficiency, while Rolls-Royce continues to rely on on-premise ERP (SAP R/3) for tighter control and complicated operational integration. Comparative analysis indicates benefits and drawbacks for both methods, while developing trends hint toward hybrid designs and AI-driven automation. As a result, a hybrid ERP solution is recommended for balancing control and scalability while lowering costs and enabling gradual cloud adoption. To summarize, picking the suitable ERP model depends on organizational size, security requirements and long-term digital strategy.

Introduction

Enterprise Resource Planning (ERP) systems are now very important for modern organizations. They serve as a platform that links business functions like finance, supply chain, human resources and production into one system. Initially, ERP systems were first used as an on-premise. According to Al Hayek and Abu Odeh (2020), “On-premise

software is installed locally on a company’s computers and servers. The implemented software features may be accessed usually through client components installed on the client’s machine”. This model indicates that the company itself is responsible for the implementation and the maintenance of the ERP systems, which often requires a lot of time.

On the other hand, the emergence of cloud-based ERP has completely changed how enterprises control their resources. Unlike traditional on-premise ERP, cloud-based ERP operates on a cloud infrastructure, enabling users to retrieve information easily through client configurations. It eliminates the need for companies to invest heavily in physical hardware or internal maintenance as all the system management is handled by the cloud vendor (Navaneethakrishnan, 2013).

This paper aims to compare cloud-based and on-premise ERP systems, analyzing their respective advantages, disadvantages and challenges. Furthermore, it discusses future directions for enterprise adoption, emphasizing future directions that enterprises can consider

Background / Literature Review

Prior to seven years or more, companies relied on a local custom Enterprise Resource Planning (ERP) (on-premise ERP) which was frequently far from business best practices. According to Al Hayek and Abu Odeh (2022), it follows business demands and does not force companies to create and apply business practices which is why the ERP system cannot maintain the company continuity. Several years ago, major ERP vendors began introducing cloud versions of their systems to replace traditional on-premise models. As highlighted by Mezghani (2019), cloud computing has the

potential to drastically change the ERP environment since a vendor can be tailored to match the needs of the user and also host the data safely somewhere on the Internet. However, there are still many research questions surrounding this evolutionary approach to ERP systems.

As discussed by Johansson et al. (2025), making the option to deploy a cloud-based ERP is frequently considered easier for smaller organizations because the decisions are not as complicated. Larger organizations who have already invested extensively in on-premise ERP infrastructure may face a more difficult decision due to their size which potentially leads to a prolonged use of the same ERP. In addition to that, on-premise solutions still provide deeper customization and complete control over data and interfaces (Al Hayek et al., 2020) which is why the organizations might decide to stay to use the same ERP. Nevertheless, cloud ERP offers better flexibility in which it reduces upfront capital expenditures while also speeding up deployment and updates according to the research study (Al Hayek et al., 2020). This is especially appealing to growing or leaning into IT departments.

Looking ahead, industry highlights hybrid ERP which is a mixture of many types of IT deployment methods including on-premises, private and public clouds (Al Hayek et al., 2020). According to Jaiswal (2025), this method is especially appropriate for large manufacturers with solid IT teams and the finances to spend in highly customized, integrated and secure ERP systems. Overall, existing literature points out the growth of cloud and hybrid ERP systems but more research is needed to analyze their practical limitations and long-term viability across various organizations.

Analysis and Discussion

How the technology works or functions

I. On-premise ERP

On-Premise ERP systems are deployed within an organization's own IT infrastructure, where the hardware, software and data are managed internally by the company's IT department. This means that the ERP systems run on the organization's local servers and network instead of being hosted by different organizations. The system architecture is made to satisfy the organization's technical and functional requirements. While the technical component uses data servers, mainframe computers, networking hardware, middleware and security mechanisms to guarantee secure system operations, the functional component concentrates on choosing appropriate EERP modules to support a range of business functions (Demiri, 2015).

Figure 1 illustrates how the On-Premise ERP systems use a client-server architecture, where users access ERP applications from their computers or devices, while all data is stored and processed on a central server. This setup typically follows a three-tier structure as shown in Figure 1, where the presentation tier provides graphical user interface, application tier handles business logic and lastly the data tier manages data storage and retrieval (Demiri, 2015)..

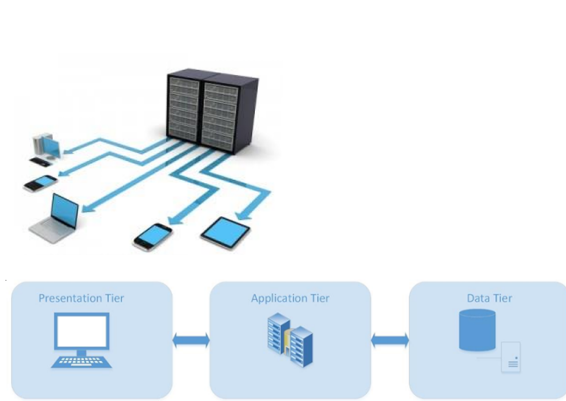


Figure 1 : On-Premise Architecture and Tier Architecture

II. Cloud-based ERP

Cloud-based systems are typically categorized into several service models and ERP implementations usually fall under Software as a Service (SaaS), as they allow end users to access applications online without managing the underlying hardware or software infrastructure (Navaneethakrishnan, 2013). These systems typically operate on a public cloud model, where services are delivered over the internet and maintained by external providers. This service-oriented structure offers software functionalities directly to users making ERP applications more accessible, flexible and cost-effective (Navaneethakrishnan, 2013).

Figure 2 illustrates how cloud-based ERP operates within a service-oriented architecture. In this model, ERP providers and experts publish ERP web services in the cloud while companies select, implement and access them through the internet.

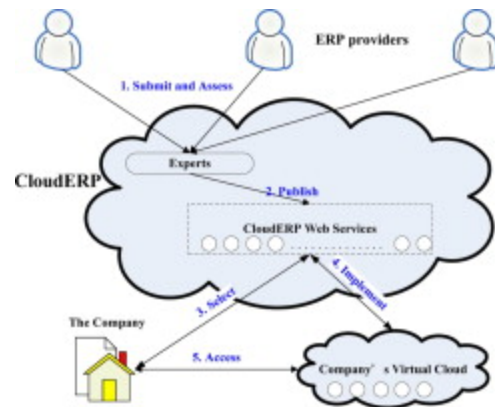


Figure 2: Architecture of Cloud ERP System

III. Comparative analysis between On-Premis and Cloud-Based ERP

On-Premise ERP and Cloud-Based ERP are different primarily in their deployment models, infrastructure management and accessibility. On-Premise ERP systems are installed and maintained within the organization itself which requires expertise in IT, giving the company full control over its data. On the other hand, Cloud-Based ERP operates under a SaaS model where applications are hosted on external cloud infrastructure and accessed through the internet, allowing users to use ERP functionalities without handling hardware and complex installations. This service-oriented model offers flexibility, scalability and cost efficiency meanwhile On-Premis ERP provides more control and customization.

Case examples and applications

Prior to 2018 according to Azhar et al. (2022), Engineering Company (EngCo) processed business-related data using the UBS accounting system which was created to document business transaction processes and provide reports. The system was difficult to import and update data resulting in information processing delays that caused uncollectible debts and delayed audited

financial reports [1]. To address the problem, EngCo made the decision to implement cloud-based ERP to enhance EngCo's business by integrating data across departments. EngCo's cloud-based ERP gives real time, remote access to data and information allowing accountants and managers to operate well from anywhere as well as increased overall productivity by decreasing delays and streamlining data changes.

Even so, as highlighted by Yusuf et al. (2004), there is still an industry like Rolls-Royce that still uses on-premise ERP. They chose the on-premise ERP software solution known as SAP R/3. Rolls-Royce formerly relied on over 1500 costly, difficult-to-maintain systems that lacked consistent and accessible data for rapid decision-making and performance assessment. Following deployment, the system provides advantages including business process automation, improved supply chain performance via e-communication and e-commerce and quick access to management information.

Advantages and disadvantages

This section examines the advantages and disadvantages of cloud-based ERP as well as on-premise ERP to highlight both benefits and potential constraints that enterprises may encounter. Table 1 includes the benefits and drawbacks of cloud-based ERP according to Abd Elmonen et al. (2016) while table 2 includes the advantages and disadvantages of on-premise ERP according to Ainapur (2014) and Hoseini (2012).

Table 1: Advantages and disadvantages of cloud-based ERP

Advantages	Disadvantages
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Low upfront costs	Limits control over Cloud ERP
High scalability	Customization and integration limitations
Rapid updates and upgrades	High security risks

Table 2: Advantages and disadvantages of on-premise ERP

Advantages	Disadvantages
Greater control and ownership	High upfront costs
Strong customization capability	Poor scalability
High security environments	Less accessibility from remote locations

Challenges and Future Trends

Table 3 shows the challenges faced by both On-Premise and Cloud-Based ERP. It highlights the key points on what each of the architecture struggles with several issues and challenges.

Table 3: Challenges of On-Premise and Cloud-Based ERP Systems

Type of ERP	Challenges
On-Premise	High installation and maintenance cost
	Lack of IT expertise
	Complexity and long time to train individuals

Cloud-Based	Need to subscribe to cloud services with regular payments
	Requires high security measure as it relies on internet
	Hidden cost (monitoring, transition and coordination)

In upcoming years, even cloud ERP systems gain prominence, On-Premise ERP solutions will continue to evolve, focusing on integrating AI capabilities to enhance operational efficiency. Organizations will invest in AI-driven tools that can analyze a huge amount of data to provide insights, improving decision making processes (SAP 2025). However, as agentic AI advances, Cloud-Based ERP systems are expected to become more intelligent and flexible. These systems will be able to perform complex tasks with less human interaction, changing to autonomous systems in business operation, allowing organizations to make decisions smoothly (SAP, 2025).

Recommendation

Based on the discussion above, it is recommended that organizations apply a hybrid ERP system that combines both On-Premise and Cloud-Based ERP systems. This approach allows organizations to control their data while utilizing cloud-based modules for scalability. Hybrid ERP allows companies to innovate more quickly, meet industry-specific requirements, reduce costs and gradually change to the cloud while

maintaining the fundamentals of the system (Beaver, 2023).

Conclusion

In summary, this report examined cloud-based and on-premise ERP systems by comparing their deployment models, strengths and weaknesses across various organizational contents. On-premise ERP provides improved control, greater customisation and adaptability for a highly regulated environment but it demands higher upfront expenses, longer deployment processes and expert IT assistance. In comparison, cloud-based ERP enables scalability, real-time access, decreased infrastructure expenses and speedier implementation, making it perfect for enterprises seeking agility and accessibility. While traditional on-premise systems continue to service complex business contexts that require tighter system integration, case studies show how cloud-based ERP enhanced operational performance and information flow. Emerging trends speak to a shift towards hybrid models and more usage of AI-driven insights to improve automation and decision-making. As a result, companies should explore a hybrid ERP approach that balances control, scalability and gradual cloud adoption while avoiding operational skills. Overall, lessons learnt show that ERP selection should correspond with the business strategy, data requirements and digital maturity. Future study might look into long-term sustainability, the effects of AI adoption and how security evolves in hybrid ERP ecosystems

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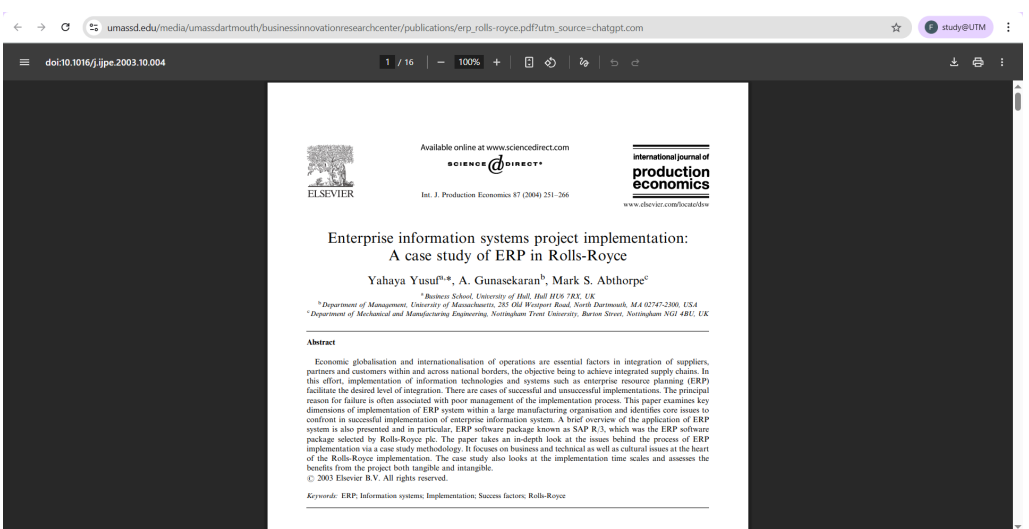
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Appendix

Logbook - <https://github.com/users/mukhriz/projects/7>

Type of AI Tools	Date	Documented Prompt & Output
ChatGPT	21/10	Prompt : Web based ERP and Cloud based ERP are the same thing or not Good question — they're related but not exactly the same. Here's the difference explained clearly 🧡 ♦ Web-Based ERP • A web-based ERP is any ERP system that can be accessed through a web browser using the internet or an intranet. • The system may still be hosted on the organization's own servers (on-premise) or on an external server. • The key feature is web access — users don't need to install software on their computers; they just use a browser. • Example: An on-premise ERP that your IT team installs on your local server but lets employees log in through a browser within the company network. ♦ Cloud ERP • A cloud ERP is a web-based ERP that is hosted on the vendor's cloud infrastructure (like AWS, Azure, or Oracle Cloud). • The vendor handles server maintenance, updates, backups, and security, while users access the system entirely online via a browser. • It is typically offered as a Software as a Service (SaaS) model — you pay a subscription instead of owning the software. ✅ In short: All Cloud ERPs are web-based, but not all web-based ERPs are cloud ERPs. • Web-based ERP → defined by <i>how you access it</i> (via browser). • Cloud ERP → defined by <i>where it's hosted</i> (in the cloud).
ChatGPT	24/10	Prompt : Suggest credible website that suggest hybrid ERP Here are two reputable websites that recommend a hybrid ERP strategy and discuss its benefits in detail: • TechTarget — "What is hybrid ERP? Benefits and use cases" explains how hybrid ERP combines on-premises core systems with cloud modules, and outlines benefits like cost optimisation, scalability, control over data, and preserving legacy customisations. TechTarget • NetSuite — "What Is Hybrid ERP and What Are Its Advantages?" lists five practical reasons to deploy hybrid ERP (faster innovation, industry/region fit, lower cost, stepping-stone to cloud, regulatory compliance). netsuite.com If you like, I can pull 5-10 more websites focusing on hybrid ERP (with summary bullets) for your thesis or project. Verification steps : - Choose website that are well known for its credibility

		
ChatGPT	22/10	can you give me at least give 5 examples of application or case examples of it from a RELIABLE RESOURCES than can be access FREE Here are 5 free-access examples (or case-studies) of ERP / on-premise ERP / ERP implementation that you can use in your report: 1. Rolls-Royce plc case study: <i>A case study of ERP in Rolls-Royce</i> — large manufacturing organisation, deep dive into technical, business and cultural issues with ERP. umassd.edu 2. Lappeenranta University of Technology multiple-case study: <i>On-premise ERP organisational post-implementation practices: Comparison between large enterprises and SMEs</i> (Hasheela) — available as PDF. Semantic Scholar 3. Free pdf: <i>Obstacles of On-Premise Enterprise Resource Planning Systems</i> by S. S. Gözükar et al. — discusses real-world on-premise ERP issues. eDepot 4. Free research paper: <i>Enterprise Resource Planning – Real blessing or a Blessing in Disguise: An Exploration of the Contextual Factors in Public Sector</i> — case examples of ERP in public sector. arXiv 5. Free article on implementation: <i>ERP case study: user adoption and implementation</i> (ERPFocus) — a real-world ERP deployment story with lessons. erpfocus.com If you like, I can pull direct download links and format them with full APA references for you. Verification steps: - Make sure the chosen points are from a reliable published article (we took the first point)

		
ChatGPT	26/10	How to write an executive summary. give a step by step format on how to write it How to write an Executive Summary — step-by-step (simple) Executive summaries must be short, crisp, and written last. Aim for 150–200 words. Follow these steps: 1. Write it last • Finish the whole report first (analysis, recommendations, conclusion), then condense. 2. Start with purpose (1 sentence) • Say what the report is about and why it was written. • Example: "This report compares cloud-based and on-premise ERP systems to guide enterprise selection for improved operational performance." 3. Summarize the context/background (1–2 sentences) • One quick sentence on the situation or problem. • Example: "Many firms face delayed reporting and high maintenance costs from legacy on-premise systems, prompting migration considerations." 4. State key findings (2–4 short sentences) • Present the most important results or insights from your analysis (rank them if useful). • Use bullet form if there are several distinct findings. • Example bullets: • "Cloud ERP reduces upfront capital expenditure and speeds deployment." Verification steps: - Do research on Google from multiple sources and take note of the similarity.